

Davide Salaorni | Ph.D. Student

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Short Bio

Davide Salaorni is a Ph.D. Student in Information Technology at the Department of Electronics, Information and Bioengineering (DEIB) of Politecnico di Milano, under the supervision of professors Marcello Restelli and Francesco Trovò.

After one year of working experience that followed the completion of my M.Sc. studies, I decided to pursue the academic path to deepen my understanding of Artificial Intelligence, particularly within the domain of Reinforcement Learning.

I am currently engaged in a research project with the Italian company RSE, wherein I am developing a Digital Twin for an Energy Storage System. This project involves integrating the framework within a Smart Grid environment and employing advanced Reinforcement Learning strategies to enhance its capabilities.

Education

PhD in Information Technology

Politecnico di Milano

Milano

Nov 2022 – Today

Ph.D. in Reinforcement Learning: algorithm design for real-world control problems.

Supervisors: Marcello Restelli, Francesco Trovò

M.Sc in Computer Science and Engineering

Politecnico di Milano

Milano

Mar 2019 – Dec 2021

Supervisors: Stefano Galelli (SUTD), Marcello Restelli (Polimi)

Thesis on optimal real-time control of water distribution systems undergoing cyber-attacks employing reinforcement learning techniques (in collaboration with Singaporean university SUTD)

B.Sc. in Computer Science and Engineering

Politecnico di Milano

Milano

Sep 2015 – Mar 2019

Job Experience

Junior Performance Consultant

Moviri S.p.A.

Milano

Oct 2021 – Nov 2022

Dynatrace Admin - Monitoring and Observability Team

Publications

Journal Articles

2025: Davide Salaorni, Federico Bianchi, Silvia Colnago, Andrea Barisione, Francesco Trovò, and Marcello Restelli. A novel digital twin for battery energy storage systems in micro-grids. *Journal of Energy Storage (under review)*, 2025. (Q1 Scimago).

2023: Andrés Murillo, Riccardo Taormina, Nils Ole Tippenhauer, **Davide Salaorni**, Robert van Dijk, Luc Jonker, Simcha Vos, Maarten Weyns, and Stefano Galelli. High-fidelity cyber and physical simulation of water distribution systems. i: Models and data. *Journal of Water Resources Planning and Management*, volume 149, page 04023009, 2023. (Q1 Scimago).

Conferences

2025: Davide Salaorni, Federico Bianchi, Francesco Trovò, and Marcello Restelli. A reinforcement learning approach for optimal control in microgrids. In *International Joint Conference on Neural Networks (under review)*, 2025. (B Core Ranking).

Workshops

2025: Davide Salaorni, Federico Bianchi, Francesco Trovò, and Marcello Restelli. Leveraging reinforcement learning for micro-grid optimal control. In *Computational Intelligence for Sustainable Energy Management in Microgrids and Renewable Energy Communities (Under Review)*. IJCNN, 2025.

2022: Davide Salaorni. Optimal real-time control of water distribution systems undergoing cyber-physical attacks. In *2nd Workshop on Control Methods for Water Resource Systems*. IFAC, 2022.

Teaching Activity

Informatica B

28 Hours

Assistant of prof. Trovò at Politecnico di Milano

Sep 2024 - Jan 2025

Teaching Programme: C, Matlab and computer science foundations.

Informatica B

26 Hours

Assistant of prof. Trovò at Politecnico di Milano

Sep 2023 - Jan 2024

Teaching Programme: C, Matlab and computer science foundations.

Talks and Seminars

AirLab RL3 PhD meetings

Deterministic Policy Gradient Methods

Politecnico di Milano

13 February 2025

RSE Academy Seminar

An introduction to Reinforcement Learning

Ricerca sui Sistemi Energetici

13 March 2024

Other Academic Activities

Reviewer Activities

International Conferences and Workshops:

- ICML - International Conference on Machine Learning
- ICLR - International Conference on Learning Representations
- IJCNN - International Joint Conference on Neural Networks

International Journals:

- IEEE - Transactions on Neural Networks and Learning Systems (Q1)
- PONE - PLoS ONE (Q1)

Participation to Summer Schools and Conferences

- EWRL - European Workshop on Reinforcement Learning (Toulouse, October 2024)
- Synapse - AI Symposium by Bending Spoons (Milano, September 2023)
- RLVS - Reinforcement Learning Virtual School (Virtual, March 2021)

Tutoring Activities

- Ph.D. Manager Full-stack Development (Polimi, July 2024 - ongoing)
- Ph.D. Portal FrontEnd Development (Polimi, Nov 2023 - May 2024)
- *What a Polimi's Ph.D. student does for us?* - Video Recording and Editing (Polimi, May 2023 - Mar 2024)
- Ph.D. Shadowing - Am I an Innovator? (Polimi, Aug 2023)

Last update: Mar 8th 2024

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